Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Problems on Taylor's and Maclaurin's series

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Taylor's and Maclaurin's series

1 Problems on Taylor's and Maclaurin's series

Study focus: Problems on Taylor's and Maclaurin's series. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000178 · EV0000179

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Alright, now let's try some problems on Taylors and McLaren series.

Key details

- As I was mentioning in the lecture, we are not going to solve problems on the Taylors and McLaren series for a function of single variable.
- But we'll try.
- And find out the Taylor series and the McLaren series for a function of two independent variables.
- So the first question is expand e power x cos y in powers of x minus 1 and y minus 4 up to second degree term.
- So the point is 1 comma pi by 4 because we want to express it in that form.
- x minus 1 into f x of 1 comma pi by four plus y minus pi by four into f y of 1 comma 5 by 4.
- So this is only the first degree term.
- The question says up to the second degree term.
- So the next one will be plus x minus 1 the whole square.
- And then plus etc.
- Of course, it's an infinite series.
- But what we are required to find is I need to calculate what is the value of the function.
- The function is e power x into cos y.
- And then we will find all the partial derivatives of this, or till whatever we require and the corresponding values.
- F of 1 is e and a cos of pi by 4 is root 2.
- Then what is f x of x y?
- F x of x y is you have to treat cos y as constant.

- What will be f y of x y now?
- I need to differentiate cos y by treating x as constant.
- Now we are done with the first order partial derivatives.
- F xx of x comma y will be I need to differentiate this again with respect to x.
- I need to differentiate fx with respect to y.
- And the last one that I need is f y f y of x comma y.
- So that means I need to differentiate f y with respect to y again.
- If you observe carefully, all these values are same and only thing they differ is because of the sign.

Example or application from the lecture

- So first let's write out what is the Taylor series expansion.
- And then now let's find the mixed second order partial derivative fxy of x y.
- That's because of the point that they have given about which we need to expand the function.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 05:00–09:59

I can write f of x comma y as f of 1 comma pi by 4 is e pi root 2.

Key details

- So in fact I can take out e by root two common from all the terms because every term here has e by root 2.
- Next one is x minus 1.
- Plus, for y it is minus, so y minus y by four.
- This is the second degree term.
- Plus I have x minus one the whole square by two factorial.
- F x is positive.
- and minus x minus one of course into two into y minus pi by four divided by two factorial and the last term is also minus I'll get y minus pi by four the whole square divided by two factorial.
- So I can stop here for now because they have asked me to expand only up till the second degree term.
- So maybe in the next problem we'll see an expansion for Taylor series or McLaren series wherein we go up to third degree term.
- F of x y is equal to log of two x plus y plus one about the point zero comma zero up to three terms.
- That's a question.
- F x at zero coma zero by two factorial plus two x y.
- f x y 0 comma zero by two factorial plus f y at zero comma zero into y square by two factorial and then of course plus etcetera other terms will continue.
- I'll get log of 1, which is 0 for me.

- So first term, the constant term is not there for this particular function.
- f x will be 1 by 2x plus y plus 1 into 2.
- Because I am differentiating with respect to x, I need to treat the other two variables as constant.
- What will be FY of X, comma Y.
- If y is again 1 by 2x plus y plus 1 and the derivative of y is just 1.
- So f y of x comma y or 0 will be just 1 because these two terms goes to 0.
- So that means f x at 0 comma 0 will be minus 4.
- But let me find the two more terms, which is fx y.
- of x y so that I can complete all the second degree term.
- And the last one would be to calculate f y of x y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 10:00–14:59

Will be the first term is 0.

Key details

- Okay, that's the first term.
- Minus 4x square by 2 factorial.
- Minus 2xy by 2 factorial.
- Actually these two terms were not necessary because the question says up to three terms.
- Two x plus y plus one will be two x minus or plus y minus two x square, okay, minus x y minus y square by two plus x.
- So the next problem we'll see it as an application of Taylor series.
- Now in the next problem we'll see how to apply this Taylor series to evaluate a value.
- This is the question, right?
- So that means just observe these values.
- Because I want the approximate value, can I say that f of 0.9 and minus 1.2 is approximately equal to f of 1 comma minus 1.
- And then so f x at 1 comma minus 1 will be you substitute y as minus 1 and this is square which is 2.
- Next is we'll find out what is f y of x comma y.
- But for now I think because we have three non zero terms, this should be sufficient.
- That means we got tan inverse of x y.
- I can write tan inverse of x y as minus pi by four plus I'm expanding about the point x minus one so x minus one into minus half plus y plus one into plus half.
- Plus, etcetera of course.
- But this also can give a better close approximation of this.

Example or application from the lecture

- Let's try another problem on find Taylor series expansion of f of x y equal to tan inverse of x y and find the approximate value of f of zero point nine comma minus one point two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 15:00–19:59

0.9 and minus 1.2 will be minus 5 by 4.

Key details

- And if you substitute here for x is 0.9, minus 1 will be just 0.1 with a minus into minus half.
- Plus y is minus one point two plus one will be again minus zero point one minus zero point two in fact.
- Minus 0.1 by 2 is just the 1 by 20.
- This is just an application problem of how we can use the Taylor series to evaluate a function at a particular point.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Taylor's and Maclaurin's series transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Taylor's and Maclaurin's series transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

C Revision and retrieval practice